

SYED MUHAMMAD USMAN ZAFAR

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

EDUCATION

National University of Computer & Emerging Sciences (FAST-NUCES), Lahore

2022 – 2026

Bachelor of Science in Computer Science (June 2026)

EXPERIENCE

ARCOMPSOL | Backend Engineer

Nov 2025 – Jul 2026

Lahore, Pakistan

- Developed a data pipeline for an enterprise insurance-tech system, achieving **>90% accuracy** in automated insurance file processing.
- Architected a dynamic metadata and RAG-based mapping system, improving auto-mapping accuracy by **30%** and reducing processing failures by **60%** through Pandas optimization.
- Rewrote the caching architecture for a high-throughput fintech product, implementing layered caching and tuned TTL strategies to cut database load, minimize API latency, and ensure high availability for critical financial transactions.
- Drove a major architectural migration of an enterprise health-tech application, moving legacy Laravel and Node.js services to a modern Next.js and NestJS stack for improved scalability and performance.
- Technologies: Python, Django, Django REST Framework, PostgreSQL, Pandas, Next.js, NestJS.

FAST School of Computing (FAST-NUCES) | AI Systems Intern & Team Lead

Jun – Aug 2025

Lahore, Pakistan

- Led development of PaperPilot, a neurosymbolic AI platform for automated academic paper co-authoring, serving **50+ beta users**.
- Designed and implemented six specialized agents (LLM Extraction, Research, Writing, Citation, Knowledge Graph Builder, and Orchestrator) coordinating the full multi-agent workflow.
- Automated **80%** of the end-to-end research workflow, reducing manual effort by **90%** and improving citation accuracy by **70%**.
- Shipped dynamic knowledge-graph visualization, auto-citation planning, and DOCX export.
- Technologies: Python, LangChain, LLMs, Multi-Agent Systems, Knowledge Graphs, NetworkX, Streamlit.

Omdena (Client: IREX) | Junior Machine Learning Engineer

Jun – Aug 2024

Remote | U.S.-based client (IREX, Washington, D.C.)

- Built an AI-powered content moderation and fact-checking tool with a global team, detecting misinformation and offensive language across Colombia and the Dominican Republic.
- Analyzed **10,000+ texts**, achieving **85%+ categorization accuracy** with advanced NLP techniques.
- Engineered and optimized the fact-classification agent, improving workflow efficiency by **30–40%**.
- Technologies: Python, Pandas, Scikit-learn, Machine Learning, Natural Language Processing.

PROJECTS

GYNAI | AI-Driven Maternal Healthcare Decision Support System (Final Year Project)

2025 – 2026

- Built a cross-platform clinical system for gynecologists and expecting mothers with React Native and FastAPI, featuring automated antenatal records, appointment booking, and wellness tracking.
- Engineered a neurosymbolic AI assistant on Neo4j, reducing multi-hop query latency by **40%** and ensuring **100% record-based** accuracy via Gemini-generated Cypher.
- Deployed a delivery-mode prediction model (C-section vs. natural) as a microservice on AWS EC2, integrated into the clinician dashboard for real-time risk assessment.
- Technologies: React Native, TypeScript, Python, FastAPI, PostgreSQL, Neo4j, AWS EC2.

Patient Influx Analysis Dashboard | Mansoorah Hospital

2025

- Developed a Python analytics dashboard covering **31 departments**, reducing manual reporting time by **50%** and enabling decision-ready insights (internal, confidential deployment).
- Technologies: Python, Streamlit, Pandas, Plotly, NumPy.

ZakatFlow | Automated Zakat Computation System

2024

- Built and deployed an automated Zakat calculation system with asset visualization, reducing manual effort by **90%** and serving **hundreds of users**.
- Technologies: Python, Streamlit, Pandas, NumPy.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, Kotlin, SQL, C++

Backend & APIs: Django, Django REST Framework, FastAPI, NestJS, Next.js, REST APIs

Data & AI: Pandas, NumPy, Scikit-learn, Machine Learning, NLP, LLMs, LangChain, RAG, Multi-Agent Systems, Knowledge Graphs, NetworkX

Databases: PostgreSQL, MongoDB, Neo4j, Firebase

Tools & Platforms: Git/GitHub, AWS EC2, Linux/Unix, Postman, Streamlit, Android Studio